

Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

2. Speech and Channel Coding Fundamentals

4 hrs · asked in **22 of 22 papers** · 5 syllabus sub-topics · a few worked examples → Ch 6 numerical companion.

Read this first:

- **Speech coding is the single most reliable topic in the whole paper.** Some part of it appears in **21 of 22 papers**, and **characteristics of the speech signal** **TOP 7/22** plus **vocoders** **TOP 10/22** plus **LPC** **TOP 8/22** are the three that recur.
 - Almost every speech question is a **pair**: characteristics of speech **plus** one coder. So learn the ACF / PDF / PSD trio once, then attach whichever coder is asked.
 - **Channel coding** is examined much more lightly, in **HOT 9/22** papers but never for more than **5**. Viterbi **HOT 4/22** and turbo **5** are the recurring ones.
 - **The one link to remember between the two halves:** speech coding **removes** redundancy to save bandwidth, and channel coding then **adds** redundancy back to survive the channel. They pull in opposite directions on purpose.
-

6.1 Speech coding: why, and the two families

Why do we need speech coding techniques? **4** **2**

70 Ma, 78 Ch · the framing paragraph for every other question in this section

Goals of speech coding

- Remove **redundancy** from the speech signal.
- Achieve **compression**: use the minimum number of bits while maintaining quality.
- Reduce **transmission and storage cost**.
- In wireless specifically: **transmit speech at the highest possible quality using the least possible channel capacity**, so more speech channels fit in a given bandwidth.

The trade-offs

- Higher speech bit rate → higher quality, but greater bandwidth and storage.
- So a balance is struck between **bit-rate efficiency** and **algorithmic complexity**.
- A third axis matters in mobile: **robustness to channel errors**. A very low rate coder throws away so much that a single bit error is audible.

The two families (state this split before naming any coder)

- **Waveform coders**
 - No speech model is required. Work on any input analog signal.
 - Designed to be **source independent**, so they code music and tones equally well.
 - Aim: reproduce the **time waveform** as closely as possible, minimising the bits per sample.
 - (+) Robust across a wide range of speakers and noisy environments. **Minimal complexity**.
 - (–) Only **moderate** bit-rate economy, ie **low compression ratio**.
 - Eg **PCM, DPCM, ADPCM, delta modulation, CVSDM**, and the frequency-domain coders below.
- **Source coders (vocoders)**
 - Use **a priori knowledge** of how speech is produced.
 - Recreate the **spectral characteristics** of speech by modelling the vocal tract as a **time-varying digital filter**.
 - Filter parameters are extracted at the encoder, **only the parameters are sent**, and the decoder **re-synthesises** speech from them.
 - (+) **Very high economy in bit rate**, down to 2.4 kbps.
 - (–) **High complexity**, and quality is low and **mechanical sounding**, though intelligible. Performance is also **talker dependent**.
- **Hybrid coders** sit between the two and are what modern cellular actually uses. verified/added

• The hierarchy of speech coding

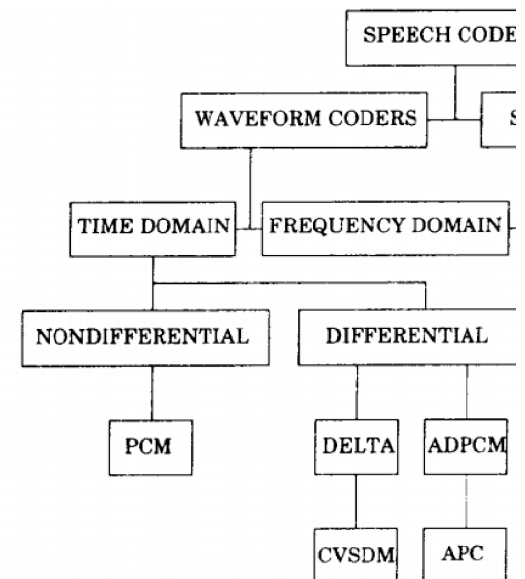


Figure 7.1

The hierarchy of speech coders. **Draw this tree and the “types of speech coder” mark is banded.**

Explain the characteristics of the speech signal, and how they are used in designing coders 8 6 4 3

TOP 7/22 72 Ash, 77 Ch, 73 Ma, 79 Bh, 80 Ch, 82 Bh, 81 Ba · the most repeated question in the chapter, and it pairs with almost every other one

Speech waveforms have properties that can be **exploited** when designing efficient coders. Three are used most often. For each one, always say **what the property is** and then **which coder design it justifies**: that second half is where the marks are.

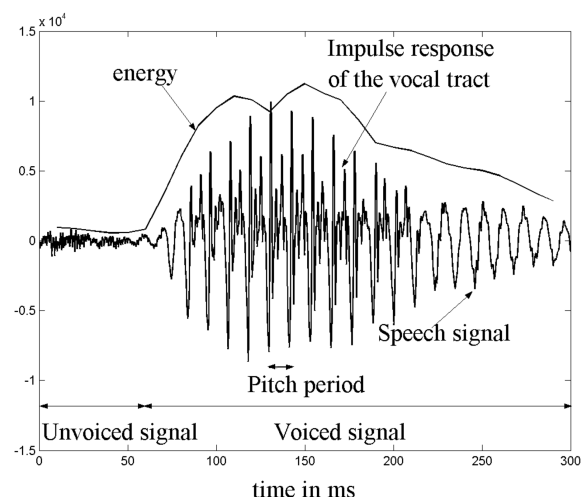
Property	What it says	What it justifies
Autocorrelation function (ACF)	Adjacent samples of a speech segment are strongly correlated . Typical adjacent-sample correlation is 0.85 to 0.9 . So every sample is largely predictable from previous samples, plus a small random error.	All differential and predictive coding schemes : DPCM, ADPCM, delta modulation, and LPC . Send the small prediction error instead of the sample.
Probability density function (PDF)	Speech amplitude has a highly non-uniform pdf: very high probability of near-zero amplitudes, small but significant probability of very high amplitudes, and a monotonically decreasing function in between.	Non-uniform quantizers and vector quantizers : put more quantization levels where the probability is high and fewer where it is low. This is why μ -law and A-law companding exist.
Power spectral density (PSD)	The PSD of speech is strongly non-flat : energy is concentrated at low frequencies. High-frequency components carry little energy but a lot of information .	Frequency-domain coding : code different frequency bands separately and allocate bits per band. Significant coding gain , but high bands must still be adequately represented or intelligibility collapses.

Two more properties worth a line

- **Band-limited**: speech is essentially confined to **300 Hz to 3400 Hz**, so 8 kHz sampling suffices. This is what makes digital speech possible at all. **verified/added**
- **Quasi-stationary**: speech statistics are roughly constant over **10 to 30 ms**, which is why every coder works on frames of that length. **verified/added**

Voiced Vs unvoiced speech +Fig

- **Voiced** sounds (“m”, “v”, all vowels) are produced by **quasi-periodic** vocal cord vibration. High energy, clear **pitch period**.
- **Unvoiced** sounds (“s”, “f”) are produced by turbulent air, so they look like **random noise**. Low energy, no pitch.
- **Every vocoder must transmit a voiced / unvoiced decision plus the pitch period.**



6.2 Frequency domain coding of speech

Explain the types of frequency domain coding of speech. Explain sub-band coding 4 6

PIN 2/22 73 Bh, 80 Bh · 80 Bh asks for sub-band coding with transmitter and receiver

The idea

- Frequency-domain coding is a **waveform** coding method.
- Think of it as **distributing quantization noise across the signal spectrum** on purpose, rather than letting it spread uniformly.
- The speech signal is split into a set of **frequency components**, which are **quantized and encoded separately**.
- The bits allocated to each component can be **varied dynamically** and shared between bands.
- Two algorithms are examined: **sub-band coding** and **adaptive transform coding**.

Sub-band coding (SBC)

- Divides speech into **4 or 8 sub-bands** using a **bank of filters**.
- Each sub-band is **sampled at its bandpass Nyquist rate**, then encoded **separately**.
- Aim: **control and distribute the quantization noise** over the whole spectrum.
 - Noise generated in one band **stays in that band**.
 - So a low-energy band is not swamped by noise leaking from a high-energy one.
- Bits per band are allocated by **perceptual importance**: low bands carry most of the energy and get the most bits.
- Bit rate range: 9.6 kbps to 32 kbps**.
- Implementation trick**: each sub-band is **low-pass translated to zero frequency** using a scheme like single-sideband modulation. That **reduces the required sampling rate** for each band.
- Receiver**: decode each band, **interpolate** back up, band-pass translate to the original position, and **sum**.
- (+) Good quality at moderate rates, and the split is easy to implement. (–) Filter bank adds delay and complexity.

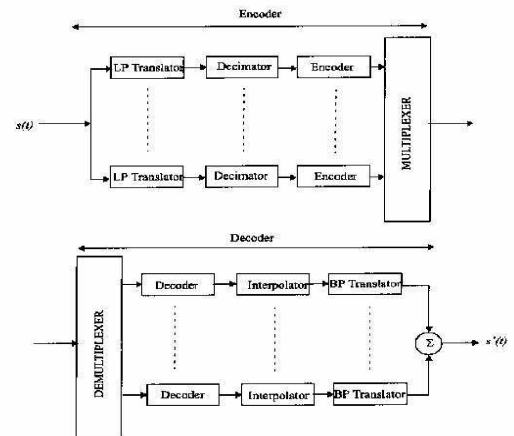


Figure 8.4 Block diagram of a sub-band coder and decoder.

Sub-band coder and decoder. LP translator, decimator and encoder per band on the left, the inverse chain on the right.

Adaptive transform coding (ATC) syllabus, no PYQ yet

- Applies a **block transform** to windowed segments of the speech waveform.
- Each segment becomes a set of **transform coefficients**, quantized separately with a number of bits **proportional to its perceptual significance**.
- Bit rate range: 9.6 kbps to 20 kbps**.
- Most common transform: the **discrete cosine transform (DCT)**

$$X_c(k) = \sum_{n=0}^{N-1} x(n) g(k) \cos \left[\frac{(2n+1)k\pi}{2N} \right], \quad k = 0, 1, \dots, N-1$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X_c(k) g(k) \cos \left[\frac{(2n+1)k\pi}{2N} \right]$$

with $g(0) = 1$ and $g(k) = \sqrt{2}$ for $k = 1, \dots, N-1$.

- SBC splits with filters, ATC splits with a transform.** That one line answers “compare them”.

6.3 Vocoders

What is a vocoder? Explain its operation with a block diagram, and the different kinds 2+6 2+3 3+5 4

6

TOP 10/22 70 Bh, 70 Ma, 71 Bh, 73 Bh, 73 Ma, 74 Bh, 74 Ma, 78 Ch, 81 Bh, 81 Ba · the single most repeated question in the whole chapter

What a vocoder does, in three steps

- **Analyze** the voice signal at the transmitter.
- **Transmit only the parameters** derived from that analysis, never the waveform.
- **Synthesize** the voice at the receiver from those parameters.
- All vocoders model **speech generation as a dynamic system** and quantify the **physical constraints** of that system.
- Those constraints then give a very compact description of the signal.
- (+) **Very high economy in transmission bit rate.**
- (–) Much more complex than a waveform coder, **less robust**, and performance is **talker dependent**.

The speech generation model +Fig (draw this first, always)

- Sound is **generated** by an excitation source, then **modulated** as it passes through the vocal tract.
- So the synthesiser needs exactly two things:
 - a sequence of **excitation signals**, and
 - a **vocal tract approximation** to modulate them.
- **Excitation** is switched by the voiced / unvoiced decision:
 - **voiced** → a **pulse source**, a series of periodic pulses at the pitch rate,
 - **unvoiced** → a **noise source**, a random sequence.
- **Vocal tract** is modelled as a **linear filter**.
- Parameters transmitted: **pitch period**, **voiced / unvoiced flag**, **pole frequencies** of the filter, **amplitude / gain**.
- **Every vocoder below is a different way of describing that filter.** Say that and the four types fall out naturally.

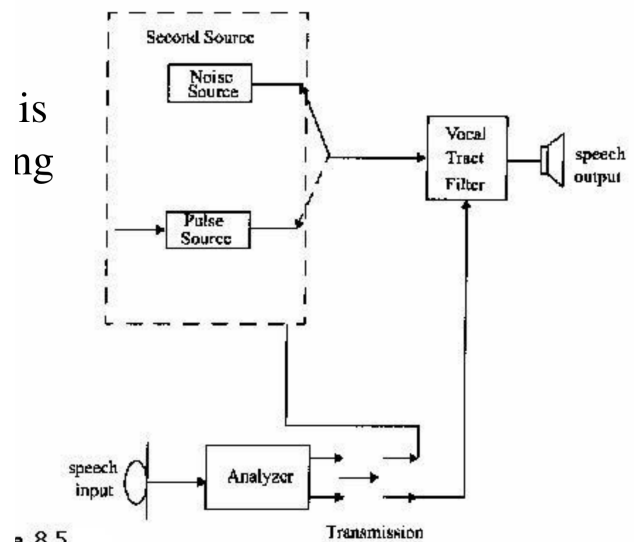


Fig 8.5
Speech generation model.

The traditional speech generation model, the basis of *all* vocoding systems. Noise source and pulse source feed a vocal tract filter.

The four classical vocoders (plus LPC)

- **Channel vocoder** = describes the filter by its **spectral envelope**, band by band.
- **Formant vocoder** = describes it by the **positions of the spectral peaks**.
- **Cepstrum vocoder** = separates source from filter by **homomorphic (log-spectrum) processing**.
- **Voice-excited vocoder** = avoids describing the excitation at all, by sending the low band as PCM.
- **Linear predictive coder (LPC)** = describes the filter by **prediction coefficients** in the **time domain**. **The most popular of all.**

Channel vocoder 4

- The **first** analysis-synthesis system ever built.
- A **frequency domain** vocoder: it determines the **envelope** of the speech signal for a number of frequency bands, then samples, encodes and multiplexes those samples.
- Also transmits, per frame, the **voiced / unvoiced decision** and the **pitch frequency**.
- Sampling is done every **10 ms to 30 ms**.

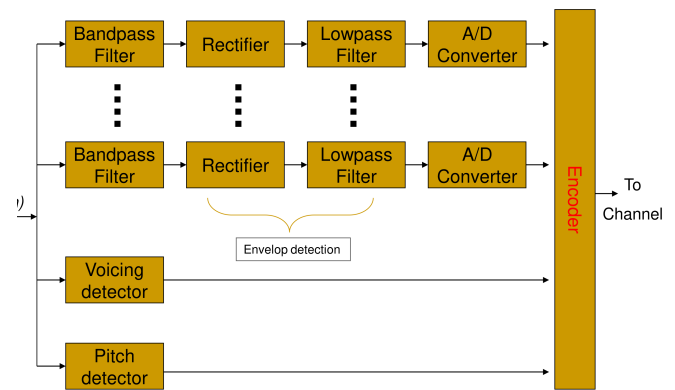
Analyzer

- A bank of **bandpass filters**, each **100 Hz to 300 Hz** wide.
- Each filter output is **rectified** then **low-pass filtered** → this pair performs **envelope detection**.
- The low-pass bandwidth is chosen to **match the rate at which the vocal tract changes**, typically a few tens of Hz.
- Each envelope is **A/D converted** and fed to the encoder.
- In parallel, a **voicing detector** and a **pitch estimator** run on the input.

Synthesizer

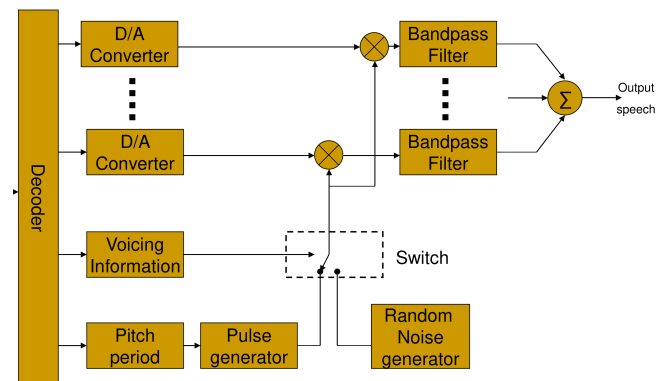
- Decoder splits the stream back into band envelopes plus voicing and pitch.
- **Switch** selects the excitation: **pulse generator** driven by the received pitch period for voiced, **random noise generator** for unvoiced.
- Excitation is **multiplied** by each band envelope, then passed through the matching **bandpass filter**.
- All bands are **summed** to give the output speech.

The Channel Vocoder (analyzer block diagram)



Channel vocoder analyzer. Bandpass filter → rectifier → lowpass filter is the envelope detector.

The Channel Vocoder (synthesizer block diagram)

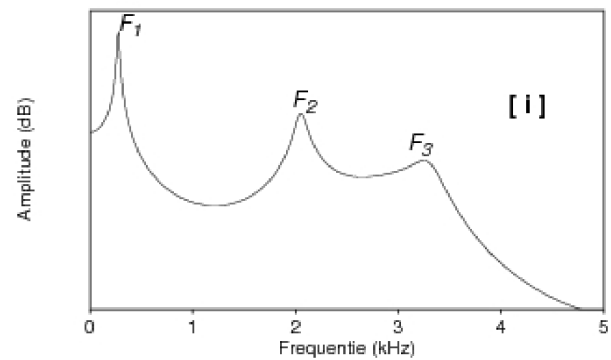


Channel vocoder synthesizer. The switch picks pulse or noise excitation from the received voicing information.

Formant vocoder 4+4

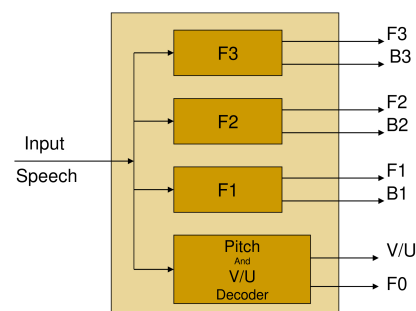
PIN 1/22 71 Bh · asked as "formant vocoder operation plus characteristics of speech"

- **Formants** = the **spectral peaks** of the sound spectrum $|P(f)|$.
 - They are frequency peaks with a **high degree of energy**.
 - Especially prominent in **vowels**.
 - Each corresponds to a **resonance of the vocal tract**.
 - Roughly **one formant every 1000 Hz**.
- Can be viewed as a **type of channel vocoder** that estimates only the **first three or four formants** in a segment of speech.
- **Key difference from the channel vocoder**: instead of sending samples of the whole power spectrum envelope, it sends only the **positions of the peaks**.
- It must also control the **intensities** of those formants, and send the **pitch period** and **voicing**.
- Per formant k it transmits F_k (the frequency) and B_k (the bandwidth).
- (+) Far fewer parameters than a channel vocoder → **much lower bit rate**.
- (-) Reliable **formant tracking is hard**, especially for unvoiced speech and for nasals, so quality suffers.



Spectral envelope of an [i] spoken by a male speaker. F_1 , F_2 and F_3 are the first three formants.

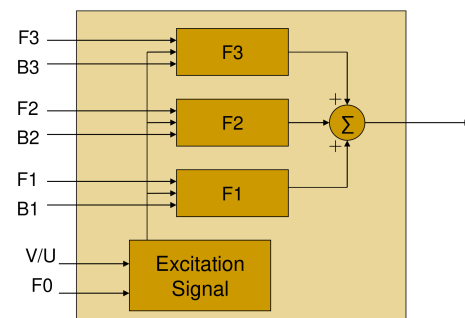
THE FORMANT VOCODER (analyzer block diagram) •



F_k : The frequency of the k th formant
 B_k : The bandwidth of the k th formant

Formant vocoder **analyzer**: extracts $F_1..F_3$, their bandwidths $B_1..B_3$, plus the voiced/unvoiced flag and pitch F_0 .

THE FORMANT VOCODER (synthesizer block diagram) •



Formant vocoder **synthesizer**: the excitation signal is shaped by three resonators and summed.

Cepstrum vocoder

- Speech is a **convolution** of an excitation source and the vocal tract system. Separating them normally needs deconvolution.
- **Objective of cepstral analysis:** separate the two **without any prior knowledge of either**.
- **Cepstrum** = the **inverse Fourier transform of the logarithm** of the estimated spectrum.
- The trick is that the log turns a product into a sum:

$$X(\omega) = E(\omega) V(\omega) \Rightarrow \log X(\omega) = \log E(\omega) + \log V(\omega)$$

- After the inverse transform the two land in **different regions**:
 - **low-frequency (low-quefreny)** coefficients → the **vocal tract spectral envelope**,
 - **high-frequency** coefficients → the **excitation**, a periodic pulse train at multiples of the pitch period.
- So they can simply be **separated by windowing** (liftering).
- **Transmit only the vocal tract cepstral coefficients**.
- At the receiver: Fourier transform them to get the **vocal tract impulse response**, then **convolve** that with a synthetic excitation to reconstruct the speech.

Voice-excited vocoder

- **Hybrid:** PCM transmission of the **lower** frequency band of speech, combined with **channel vocoding** of the **higher** bands.
- At the synthesizer a **pitch signal is generated** from the received baseband by **rectifying, bandpass filtering and clipping** it. That creates a **spectrally flattened** signal with energy at the pitch harmonics.
- **(+) This eliminates the need for pitch extraction and voicing detection**, which are the two most error-prone operations in any vocoder.
- **(–) Costs bit rate:** designed for **7.2 kbps to 9.6 kbps**, well above a pure vocoder.

Comparing the four

Vocoder	What it sends	Domain
Channel	Envelope per band, pitch, voicing	Frequency
Formant	3–4 formant frequencies + bandwidths, pitch, voicing	Frequency
Cepstrum	Low-quefreny cepstral coefficients	Cepstral
Voice-excited	Baseband as PCM + high bands as channel vocoder	Hybrid
LPC	Prediction coefficients, gain, pitch, voicing	Time

6.4 Linear predictive coders

Explain the linear predictive coder (LPC) with a neat block diagram 2+6 3+5 6

TOP 8/22 80 Ba, 81 Ba, 72 Ma, 79 Bh, 73 Ma, 76 Bh, 75 Bh, 71 Ma · almost always paired with “characteristics of the speech signal”

What makes LPC different

- The **time domain** class of vocoders. All the others above work in frequency.
- Extracts the significant features of speech **from the time waveform** directly.
- **Computationally intensive**, but **by far the most popular of the low bit rate vocoders**.
- Transmits **good quality voice at 4.8 kbps**, and intelligible speech at 2.4 kbps.

The model

- Models the vocal tract as an **all-pole linear filter**:

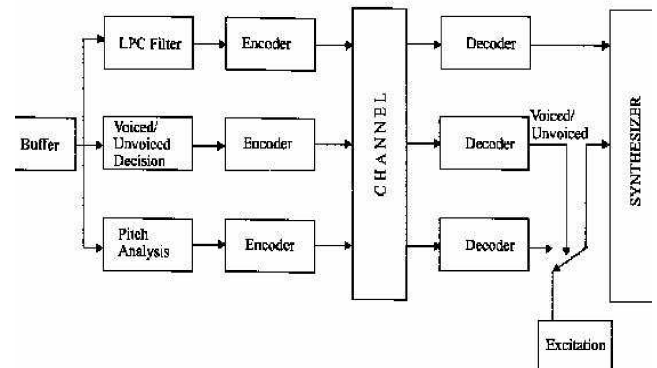
$$H(z) = \frac{G}{1 + \sum_{k=1}^M b_k z^{-k}}$$

G = filter gain, b_k = filter (prediction) coefficients, M = order, typically 10.

- **All-pole is enough** because the vocal tract is a tube with resonances, and resonances are poles. **That is the physical justification the examiner wants.**
- Coefficients are found by **minimum mean square error (MMSE)**: minimise the average energy in the **error signal**, the difference between **predicted** and **actual** speech amplitude.
- That works because of the **ACF property**: adjacent speech samples are 0.85 to 0.9 correlated, so a short-order predictor already explains most of the signal.

Operation

- **Encoder**: buffer a frame → run **LP analysis** for the coefficients → decide **voiced / unvoiced** → estimate **pitch** → compute **gain**.
- Speech type, pitch, filter coefficients and gain are **coded into a bit stream** and transmitted.
- **Decoder**: pick pulse or noise excitation, scale by the gain, drive $H(z)$ → **synthesized speech**.



Coding system.

Block diagram of an LPC coding system. Three parallel paths: LPC filter coefficients, voiced/unvoiced decision, pitch. The synthesizer recombines them.

Types of LPC (71 Ma asks for the types, 6)

- Plain LPC uses a **single pulse per pitch period**, which sounds **buzzy and synthetic**. The three variants below all improve the **excitation model**, and that is the only thing that separates them.
- **Multipulse excited LPC (MPE-LPC)**: uses **several pulses per frame**, with positions and amplitudes chosen by analysis-by-synthesis. No voiced / unvoiced decision is needed at all.
- **Residual excited LPC (RELPC)**: see below.
- **Code excited LPC (CELP)**: the excitation sequence is selected from a **codebook of zero-mean Gaussian sequences**, shared by encoder and decoder.
 - **Analysis by synthesis**: the encoder tries every codebook entry, synthesises speech with it, and keeps the one giving the smallest perceptually weighted error.
 - Only the **codebook index** is transmitted, which is why it is so efficient.
 - Bit rate **4800 bps**.

Residual excited LPC (RELPC)

- Improves quality **at the expense of a higher bit rate**, by computing and transmitting a **residual error**, exactly as DPCM does.
- Steps:
 - LPC model and excitation parameters are estimated from a **frame of speech**.
 - Speech is **synthesized at the transmitter** using them.
 - The synthesized speech is **subtracted from the original** to form the **residual error**.
 - The residual is **quantized, coded and transmitted** alongside the parameters.
- At the receiver: synthesize from the model, then **add the residual back**.
- The receiver therefore gets the part of the signal the model could not explain**, which is what lifts the quality above plain LPC.

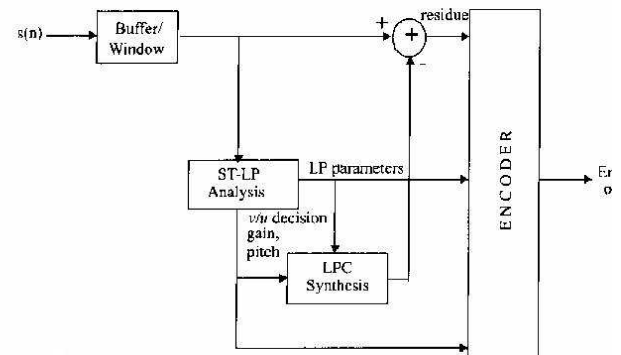


Figure 8.9
Block diagram of a RELPC encoder.

RELPC encoder. The residual is formed by subtracting locally synthesized speech from the buffered input.

The GSM codec 2

74 Ma · asked as part of a "waveform + voice coding + GSM CODEC +Fig" question

- GSM uses the **regular pulse excited, long term prediction (RPE-LTP)** codec.
- It **combines** the earlier French **baseband RELPC** codec with the German **multipulse excited long-term prediction (MPE-LTP)** codec.
- So it is a hybrid**: a source model (LPC) driving a partly waveform-coded excitation.
- Chain, left to right:
 - Pre-processing**: Hamming window, segmentation into 20 ms frames, pre-emphasis.
 - STP** (short term prediction): LPC inverse filter and STP analysis filter produce the residual e_n and the **LAR(k)** log-area-ratio coefficients.
 - LTP** (long term prediction): removes the **pitch periodicity** left in the residual, giving lag P_n and gain g_n .
 - RPE**: low-pass filter, then **grid selection** picks the best regular pulse sequence x_m .
 - All parameters are **multiplexed** into the coded speech stream.
- Output rate **13 kbps** (full rate GSM), from 104 kbps of 13-bit input samples. verified/added
- (–) **Relatively complex and power hungry**.

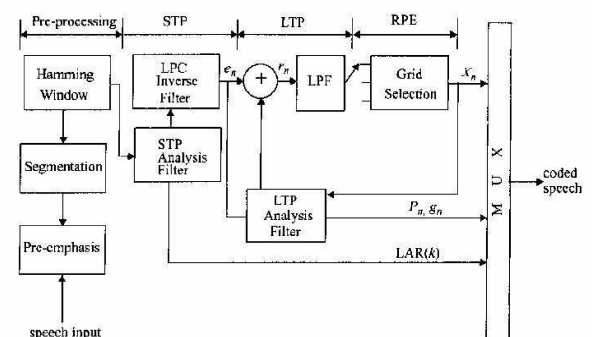


Figure 8.10
Block diagram of GSM speech encoder.

GSM RPE-LTP speech encoder: pre-processing, short term prediction, long term prediction, regular pulse excitation, multiplexer.

6.5 Channel coding

What is channel coding? Explain block codes and the definitions 2

71 Ma · the definitions below are needed by every other question in this section

The basic idea

- Add a **controlled amount of redundancy** to the message **before** it goes through the noisy channel.
- The redundancy is extra symbols added **in a known manner**, so the receiver can use them to **detect** and often **correct** errors.
- **Everyday analogy**: English already carries redundancy, which is why **"CODNG THEORY IS AN INTRSTNG SUBJECT" is still readable**.
- **Note the tension with Ch 6.1**: speech coding strips redundancy out, channel coding puts a different, engineered redundancy back in.
- Improves **small-scale link performance**: faded data can still be recovered.

Definitions (learn these five)

- A **word** is a sequence of symbols. A **code** is a set of vectors called **codewords**.
- **Hamming weight** $w(c)$ = number of **non-zero elements** in a codeword. Eg $w(10110) = 3$.
- **Hamming distance** $d(c_1, c_2)$ = number of places the two codewords **differ**. Eg $d(10110, 11011) = 3$. Note $d(c_1, c_2) = w(c_1 - c_2)$.
- **Minimum distance** d' = the smallest Hamming distance between any two codewords, $d' = \min_{i \neq j} d(c_i, c_j)$.
- **Code rate** of an (n, k) code = k/n . It is the fraction of the codeword that is actual information.

Error detection and correction capability

- To **detect** t errors per block: $d' \geq t + 1$
- To **correct** t errors per block: $d' \geq 2t + 1$
- **So minimum distance is the single number that decides how strong a code is.**

Block codes

- A **block code** is a set of **fixed length** codewords. That fixed length is the **block length** n .
- A block code of size M over a q -symbol alphabet is a set of M q -ary sequences of length n .
- If $q = 2$ the symbols are **bits** and the code is **binary**.
- Usually $M = q^k$, and we call it an (n, k) **code**: k information symbols in, n out, so $n - k$ **redundant symbols**.
- Block codes are **forward error correction (FEC)** codes: no return channel is needed.

Linear block codes (LBC)

- A code is **linear** if:
 - the **sum of any two codewords** is also a codeword,
 - the **all-zero codeword** is always a codeword.
- Consequence: for a linear code the **minimum distance equals the minimum weight** of any non-zero codeword, $d' = w'$. **This makes d' far cheaper to compute: check M weights instead of M^2 distances.**
- **Careful**: the presence of an all-zero codeword is **necessary but not sufficient** for linearity.
- **Matrix description**: a **generator matrix** G ($k \times n$) encodes an information word i of length k into a codeword c of length n :

$$c = iG$$

Cyclic codes

- A code C is **cyclic** if it is **linear** and **any cyclic shift** of a codeword is also a codeword.
- I.e. if $a_0 a_1 \dots a_{n-1} \in C$ then $a_{n-1} a_0 \dots a_{n-2} \in C$.
- Eg $C_1 = \{0000, 0101, 1010, 1111\}$ is cyclic. $C_2 = \{0000, 0110, 1001, 1111\}$ is **not**, though it is **equivalent** to C_1 : swapping the third and fourth components of C_2 gives C_1 .
- (+) Encoding and syndrome computation reduce to **polynomial division, which is a simple shift register**. That is why nearly every practical block code is cyclic.

Explain the 7-bit Hamming code method with a relevant example 5

PIN 1/22 80 Ch · worked both ways (encode and detect) in the Ch 6 companion

- A linear block code able to identify and correct any single-bit error within a block. Written as an (n, k) Hamming code.
- Uses modulo-2 arithmetic, ie XOR.
- Parameters, for any integer $m \geq 3$:
 - block length $n = 2^m - 1$
 - message bits $k = 2^m - m - 1$
 - parity check bits $n - k = m$
 - minimum distance $d_{\min} = 3$, so error-correcting capability $t = \lfloor (d_{\min} - 1)/2 \rfloor = 1$
 - code efficiency (rate) $= k/n$
- For $m = 3$: $n = 7$, $k = 4$. **That is the "7-bit Hamming code" the paper asks for.**
- (–) Offers **little protection against burst errors**, since it corrects only one bit per block. Interleaving (Ch 5) is the standard fix.

Structure

- Parity bits are **inserted in between** the data bits, at positions 1, 2, 4, ... (the powers of 2).
- Layout, bit 1 on the right: $D_7 D_6 D_5 P_4 D_3 P_2 P_1$
- Each parity bit covers the positions whose binary index includes its own bit:
 - P_1 covers positions **1, 3, 5, 7**
 - P_2 covers positions **2, 3, 6, 7**
 - P_4 covers positions **4, 5, 6, 7**

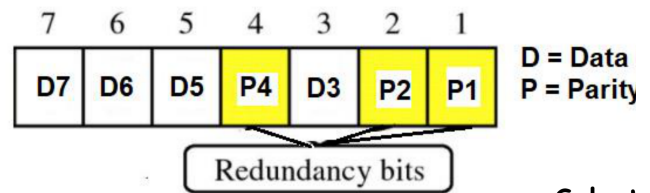


Fig : 7 bit Hamming code

$P_1 =$

Worked example 1: encode 1011 with even parity

- Data goes into positions 7, 6, 5, 3: $D_7=1, D_6=0, D_5=1, D_3=1$.
- P_1 = parity of positions 1,3,5,7 = $(P_1, 1, 1, 1)$. Three ones already, so $P_1 = 1$ to make it even.
- P_2 = parity of positions 2,3,6,7 = $(P_2, 1, 0, 1)$. Two ones, so $P_2 = 0$.
- P_4 = parity of positions 4,5,6,7 = $(P_4, 1, 0, 1)$. Two ones, so $P_4 = 0$.
- Transmitted codeword (bit 7 first): **1 0 1 0 1 0 1**

Worked example 2: check the received word 1011011

- P_1 group (positions 1,3,5,7) reads **1011** → odd parity → error, set $P_1 = 1$.
- P_2 group (positions 2,3,6,7) reads **1001** → even parity → no error, set $P_2 = 0$.
- P_4 group (positions 4,5,6,7) reads **1101** → odd parity → error, set $P_4 = 1$.
- **Syndrome** = $P_4 P_2 P_1 = 101 = 5$ in decimal.
- **So the received word is incorrect, and bit position 5 is the one in error.** Flip it to correct the word.
- **That is the whole elegance of the Hamming code:** the syndrome read as a binary number is the position of the faulty bit.

Other block codes syllabus, no PYQ yet

- **Hadamard codes**
 - Codewords are the **rows of a Hadamard matrix** A : an $N \times N$ matrix of 1s and 0s in which every row differs from every other in **exactly** $N/2$ places.
 - One row is all zeros; the rest have $N/2$ zeros and $N/2$ ones.
 - Minimum distance $d_{\min} = N/2$, so it is a **very strong but very low rate** code.
 - Used for the **Walsh functions** that separate IS-95 CDMA channels.
- **Golay codes**
 - Linear binary **(23,12)** codes with $d_{\min} = 7$, so they correct **3 bits**.
 - **The only non-trivial perfect code that exists.** Every possible received word lies within distance 3 of exactly one codeword, so maximum likelihood decoding is exact.
- **BCH (Bose–Chaudhuri–Hocquenghem) codes**
 - **One of the most powerful known classes of linear cyclic block codes.**
 - A **subset of cyclic codes** whose generator polynomial roots are chosen carefully to give good error-correcting capability.
 - Each BCH code corrects up to t **random errors** per codeword.
 - Parameters: block length $n = 2^m - 1$ ($m \geq 3$), message bits $k \geq n - mt$, minimum distance $d_{\min} \geq 2t + 1$.
- **Reed–Solomon (RS) codes**
 - A subclass of **non-binary** BCH codes.
 - **The encoder operates on multi-bit symbols, not individual bits.**
 - For m -bit symbols: block length $n = 2^m - 1$ symbols, message k symbols, parity $n - k = 2t$ symbols, $d_{\min} = 2t + 1$.
 - (+) Because a whole symbol counts as one error however many of its bits are wrong, RS codes are **excellent against burst errors**.
 - Used in CD / DVD, deep space links, and digital broadcasting.

Explain convolutional encoding and decoding, and the Viterbi algorithm 5 3

HOT 5/22 74 Bh, 75 Bh, 70 Ma, 72 Ash, 73 Bh · **Viterbi alone is asked in 4 papers** → worked example in the Ch 6 companion

How convolutional codes differ from block codes

- In a block code, each codeword depends **only** on the current block of k message digits.
- In a convolutional code, the block of n **code digits** produced at each time unit depends on the current k message digits **and on the previous ones held in memory**.
- **So the encoder has memory. That is the whole difference.**
- No fixed block boundary is needed, which suits a **continuous** data stream.

Encoder parameters + Fig

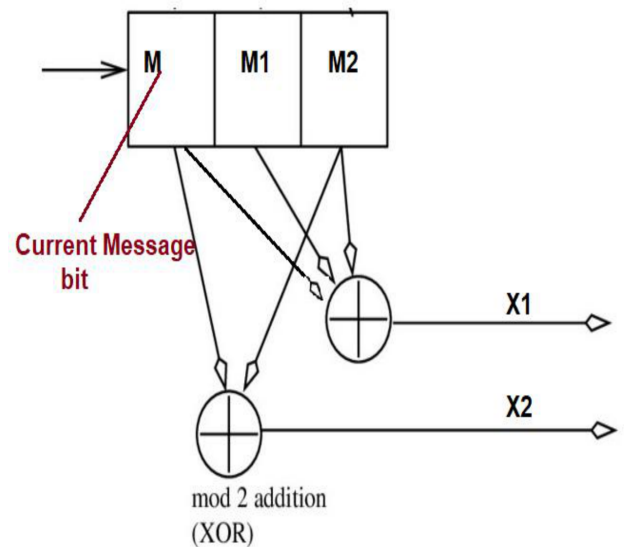
- **Constraint length** K = number of symbols the shift register can store, ie how far back the memory reaches. For the figure, $K = 3$.
- If the register stores m previous information frames of length k_0 , then $v = mk_0$.
- $k = 1$ message bit in, $n = 2$ encoded bits out.
- **Code rate** $= k/n = 1/2$, **code dimension** $= (n, k) = (2, 1)$.
- Outputs are **mod-2 sums (XOR)** of selected register stages:

$$X_1 = M + M_1 + M_2, \quad X_2 = M + M_2$$

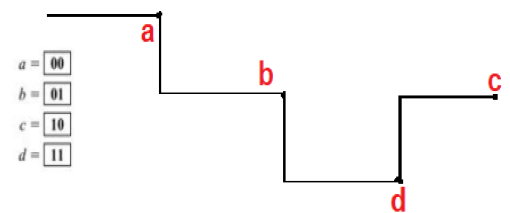
- **The tap pattern is the code.** Two encoders with the same K but different taps are different codes with different performance.

Three ways to represent the same encoder

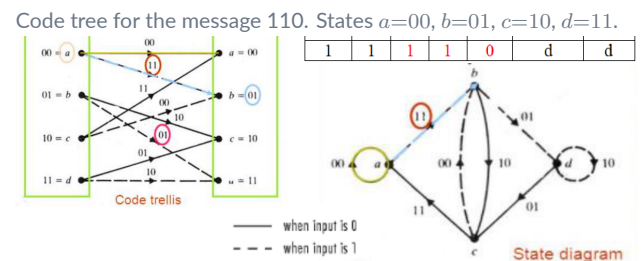
- **Code tree:** every branch is one input symbol, labelled with the output pair. **Input 0 takes the upper branch, input 1 the lower.** **(-) Grows exponentially**, so it is unusable beyond a few bits.
- **State diagram:** nodes are the register contents, arcs are transitions. Compact, but hides time.
- **Trellis:** the state diagram **drawn against time**. **A compact representation of the code tree:** the tree's repeating structure is folded back onto the same set of states at every step. This is what the decoder actually walks.



Rate 1/2, $K = 3$ convolutional encoder. Two mod-2 adders tap the shift register.



as for 110 message



Code trellis and the equivalent state diagram. Solid line = input 0, dashed = input 1.

Viterbi decoding 3 5

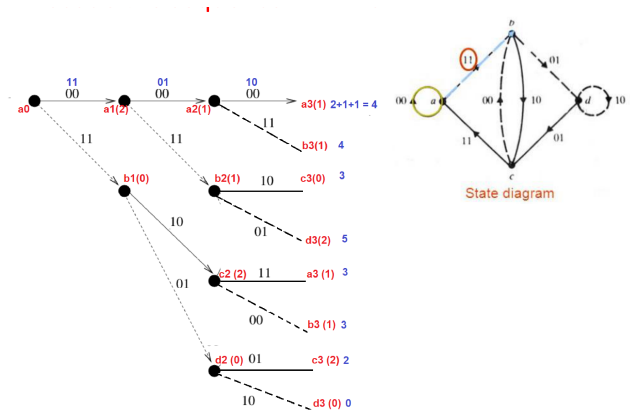
- Used for decoding convolutional codes. Also known as **maximum likelihood decoding**, because it finds the **single most likely transmitted sequence**, not the most likely symbol.
- It is the practical implementation of **MLSE** (Ch 5).

The algorithm

- Walk the **trellis** one time step at a time.
- At each step, every node has **two incoming branches** (for a rate 1/2 code).
- For each branch compute a **metric**: the **Hamming distance** between the branch's expected output and the actually received pair. (Soft-decision decoders use Euclidean distance instead.)
- Add the branch metric to the surviving path metric that led into it.
- Keep only the smaller of the two, and discard the other.** The kept one is the **survivor** for that node.
- Repeat for every node at every step. **The number of survivors never grows**, which is exactly why the algorithm is tractable while an exhaustive tree search is not.
- At the end, trace back the survivor with the lowest total metric. Its branch labels give the decoded message.

Why it is optimal

- Discarding the worse of two paths into a node is safe: **any future extension of the worse path can be applied equally to the better one**, so the worse path can never win later.
- Complexity is **linear in the message length** but **exponential in the constraint length K** , which is why practical K stays around 7 to 9.



Decoding $Y = 11\ 01\ 10$. Path metrics accumulate along the trellis; at each node the larger metric is discarded.

Define turbo coding 2 5

PIN 2/22 82 Bh, 74 Ma

- A turbo code **combines the capability of convolutional codes with channel estimation theory**. It can be thought of as **nested or parallel convolutional codes**.
- **Structure** +Fig: two recursive systematic convolutional encoders C_1 and C_2 , working on the **same input bits** but in **different orders**.
 - M in the figure is a **memory register**.
 - A **delay line** and an **interleaver** (which reorders the information bits) force the input d_k to appear in different sequences at the two encoders.
 - **The interleaver is what makes it a turbo code**. Without it the two encoders would see the same bit pattern and fail on the same errors.
 - Because the encoders are **systematic**, the input sequence d_k appears **directly at the output** x_k alongside the parity streams y_{1k} and y_{2k} .
- **Decoding is iterative**: two soft-input soft-output decoders exchange **extrinsic information** and refine each other's estimates over n_1 and n_2 iterations. The name comes from this feedback, by analogy with a turbocharger. verified/added
- **Why it matters**: coding gains **far superior** to all previous error correcting codes, letting a wireless link come **amazingly close to the Shannon capacity bound**.
- (–) **Large decoding delay** from the interleaver and the iterations, so it suits data far better than real-time voice.

Coding gain syllabus, no PYQ yet

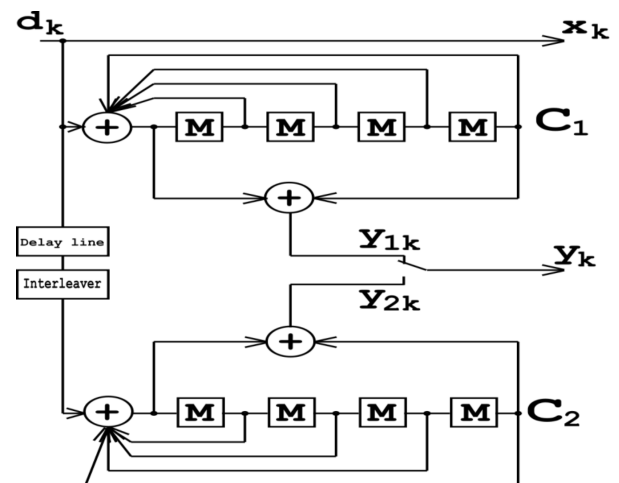
- For a given modulation and target BER, **coding gain** = the ratio of the **SNR per bit required uncoded** to the **SNR per bit required coded**:

$$G(\text{dB}) = \left(\frac{E_b}{N_0} \right)_{\text{uncoded}} (\text{dB}) - \left(\frac{E_b}{N_0} \right)_{\text{coded}} (\text{dB})$$

- Ie **how much transmit power the code saves you**.
- Every error control code has its own coding gain, depending on the code, the decoder implementation and the channel BER.

Trellis coded modulation (TCM) syllabus, no PYQ yet

- Traditionally coding and modulation are designed **separately**, even though both exist to fight channel errors.
- TCM **integrates the channel encoder with the modulator**, so coding gain is obtained **without any bandwidth expansion**.
- Uses **signal set expansion** (eg 8-PSK instead of 4-PSK) to supply the redundancy, then designs the coding and the signal mapping **jointly** to maximise the **free distance**, ie the minimum Euclidean distance between coded sequences.
- Decoded by a **soft decision maximum likelihood sequence decoder**, ie Viterbi again.
- Gains as large as **6 dB** with **no bandwidth expansion** and **no reduction in information rate**.



Turbo encoder. Two systematic convolutional encoders C_1 and C_2 ; the interleaver ahead of C_2 makes the two see different bit orders.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELTP	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise

Short	Means	Short	Means
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK,	differential (binary) PSK
		DBPSK	
CPFSK	continuous phase FSK	MFSK,	M-ary FSK, M-ary PSK
		MPSK	
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB, VSB	double, single, vestigial sideband
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	<i>BT</i>	GMSK filter bandwidth × symbol time
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
SSMA	spread spectrum multiple access	SDMA	space division multiple access
FHMA	frequency hopped multiple access	DSMA	direct sequence multiple access (= CDMA)
FCDMA	hybrid FDMA/CDMA	DS/FHMA	hybrid direct sequence / frequency hopped
TCDMA	time division CDMA	TDFH	time division frequency hopping
IM	intermodulation (product)	MU-MIMO	multi-user MIMO
WLAN	wireless local area network	WMAN	wireless metropolitan area network
WPAN	wireless personal area network	AP	access point
BSS, IBSS	basic service set, independent (ad hoc) BSS	ESS	extended service set
CSMA/CA	carrier sense multiple access with collision avoidance	CSMA/CD	... with collision detection
DCF	distributed coordination function	PCF	point coordination function
RTS/CTS	request to send / clear to send	CCK	complementary code keying
WEP, WPA	wired equivalent privacy, WiFi protected access	U-NII	unlicensed national information infrastructure band
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
PCN	personal communication network	SMS	short message service
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet

data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

One clash to watch: **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.